

MAT627 Chapter 2.3 Programming Assignment

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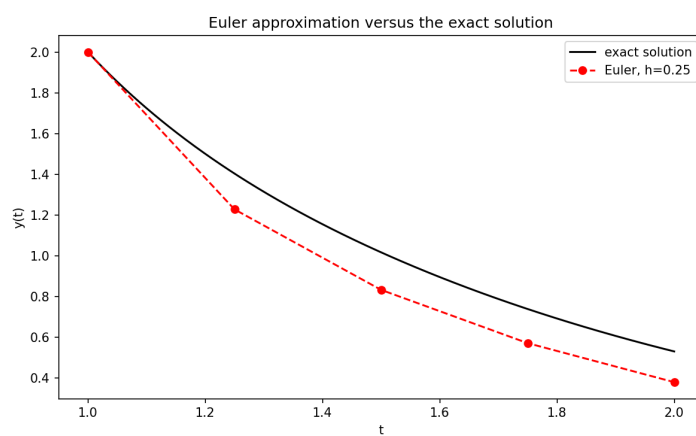


Figure 1: Part 2: Euler approximation with $h = \frac{1}{4}$ against the exact solution.

1 Part 1

2 Part 3

3 Part 4

4 Part 5

Table 1: Euler's method with $h = \frac{1}{4}$.

k	t_k	y_k	$y(t_k)$	Error = $y(t_k) - y_k$
0	1.00	2.000000×10^0	2.000000×10^0	0.000000×10^0
1	1.25	1.227324×10^0	1.403199×10^0	1.758746×10^{-1}
2	1.50	8.321502×10^{-1}	1.016410×10^0	1.842600×10^{-1}
3	1.75	5.704468×10^{-1}	7.380098×10^{-1}	1.675630×10^{-1}
4	2.00	3.788266×10^{-1}	5.296871×10^{-1}	1.508605×10^{-1}

Table 2: Part 3.

h	y_N	$E_h = y(2) - y_N$	Rate
0.5	1.829096×10^{-1}	3.467775×10^{-1}	–
0.25	3.788266×10^{-1}	1.508605×10^{-1}	1.200795
0.125	4.587866×10^{-1}	7.090049×10^{-2}	1.089347
0.0625	4.952620×10^{-1}	3.442511×10^{-2}	1.042334
0.03125	5.127189×10^{-1}	1.696823×10^{-2}	1.020625

Table 3: Part 4.

h	y_N	$E_h = y(2) - y_N$	Rate
0.5	5.315588×10^{-1}	-1.871677×10^{-3}	–
0.25	5.297856×10^{-1}	-9.846674×10^{-5}	4.248551
0.125	5.296926×10^{-1}	-5.484396×10^{-6}	4.166232
0.0625	5.296874×10^{-1}	-3.212064×10^{-7}	4.093760
0.03125	5.296871×10^{-1}	-1.940031×10^{-8}	4.049349

Table 4: Part 5.

h	y_N^r	$E_h = y(2) - y_N^r$	Rate
0.25	6.826012×10^{-1}	-1.529141×10^{-1}	–
0.125	5.566786×10^{-1}	-2.699153×10^{-2}	2.502143
0.0625	5.353216×10^{-1}	-5.634492×10^{-3}	2.260149
0.03125	5.309775×10^{-1}	-1.290431×10^{-3}	2.126433